

SMATH : peter.vlasschaert@gmail,05/02/2022

ref:youtube,https://www.youtube.com/watch?v=YG7U3ZobplE

this my first session of smath

use: create text > "

unit use : smath

$a := 5 \text{ m}$ $b := 5 \text{ ft}$ use: create unit > ' + tab
 $s := a + b$
 $s = 6.524 \text{ m}$

complex number : arithmic

$cm1 := i$ $cm2 := 3 \cdot i + 5$ $vcm := \sqrt{-1}$ $vcm = i$
 $cm := cm1 + cm2$
 $cm = 5 + 4 \cdot i$

exponential ,log,ln : arithmic

$\exp(1) = 2.7183$ $\log_e(10) = 2.3026$ $\ln(10) = 2.3026$
 $\log_{10}(100) = 2$ use $10^2 = 100$

vectors & matrices : arithmic

vectors = column matrices,T

$r := \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}^T$ $v := \begin{bmatrix} 4 & 5 & 6 \end{bmatrix}^T$

dot product

 $r \cdot v = 32$

cross product

$$r \times v = \begin{bmatrix} -3 \\ 6 \\ -3 \end{bmatrix}$$

matrices:rows & columns

$$m1 := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad m2 := m1^T$$

$$m2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

matrices:range

$$m3 := [1; 1.5 \dots 4]^T$$

$$m3 = [1 \ 1.5 \ 2 \ 2.5 \ 3 \ 3.5 \ 4]$$

$$m4 := [1 \dots 4]^T$$

$$m4 = [1 \ 2 \ 3 \ 4]$$

differentiation : symbolic, ctrl +. don't use =

first order differentiation:

$$\frac{d}{dx} \sin(x) = \cos(x) \quad \text{use : toolbar}$$

higher order differentiation:

$$\frac{d^2}{dx^2} \sin(x) = -\sin(x) \quad \text{use lower x of the differential ,qwertz ,use ; or ,}$$

integration : arithmic, use =, no symbolic use smath without maxima

$$\int_0^2 \sin(x) dx = 1.4161$$